EVALUATION OF PHASE CHANGE MATERIALS FOR THERMAL ENERGY STORAGE IN WILDLAND FIREFIGHTER PROTECTIVE SHELTERS

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Abstract

This paper presents a preliminary analysis of an emergency fire shelter intervention designed to enhance thermal buffering without exceeding wildland firefighter pack limits for mass, stowed volume, costs, and deployment ergonomics. Wildfires are increasing in frequency and burned acreage, with U.S. firefighter entrapment incidents reaching a 20-year high already in 2026 [1,2]. Wildland firefighters face increasingly severe operational hazards, including direct flame impingement, thermal entrapment during suppression operations, and radiant heat fluxes exceeding 277 kW/m^2 from fire temperatures reaching 2192°F (1200°C) [3,4].

During burn-over events, emergency fire shelters serve as the last line of defense, reflecting radiant heat with an aluminum foil-laminate outer layer; however, minimal thermal buffering is provided once heat penetrates the outer barrier. Heat accumulation inside the shelter increases occupant exposure. The life-threatening consequence of this thermal limitation was demonstrated in the 2013 Yarnell Hill Fire, where 19 firefighters lost their lives despite deploying their emergency shelters [5]. Efforts to improve shelter performance have focused on decreasing weight and adding insulation materials, leaving latent-heat buffering largely unexplored [6].

To address this challenge, this work compares organic and inorganic phase change materials (PCMs), including RT42 Paraffin, SP50 Gel, and PlusICE S46, against the M-2002 Fire Shelters [7]. Candidate PCMs were compared based on melting temperature, latent heat of fusion, thermal conductivity, density, and mass. The design integrates a PCM layer into the shelter wall that acts as a thermal battery, absorbing heat as latent heat during melting to delay interior temperature rise. The internal architecture was parameterized and optimized by evaluating layer ordering, air-gap thickness, and shelter geometry to determine how arrangement affects heat transfer.

The PCM designs were evaluated against the M-2002 shelter [7], using a transient lumped-parameter model with a 6-node baseline and 7-node PCM model and an air-gap thermal resistance. External loads include Stefan-Boltzmann radiation and forced convection. The design target is to ensure the interior temperature remains below the 122°F (49°C) design limit longer

than the current shelter design, under an exposure of 277 kW/m^2. This limit was selected because this is the highest temperature humans can survive for a short duration [8].

Each PCM shelter was evaluated at 2192 ° F (1200°C) for 30 minutes. The baseline M-2002 shelter's interior air reached 122°F (49°C) after 11.6 s. Adding a Paraffin RT42 PCM layer extended this to 62.1 s, while SP50 Gel and PlusICE S46 extended it to 98.5 s and 83.2 s, respectively. The Paraffin fully melted after 33.7 s, followed by rapid interior heating, demonstrating the limitation of finite latent heat capacity. Adding Paraffin RT42 PCM provided an additional mass of 4.66 kg, PlusICE S46 added 9.05 kg, and SP50 Gel added 8.43 kg. The additional mass required to extend this protection establishes a trade-off between extended thermal endurance and mass limits, demonstrating that PCM integration significantly increases life-safety protection time relative to standard shelters under peak thermal loads. Future work will assess the designs under varied exposures and investigate different geometric and layer arrangements to minimize added mass and maximize thermal protection.

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